

How many outputs will a decimal-to-BCD encoder have?

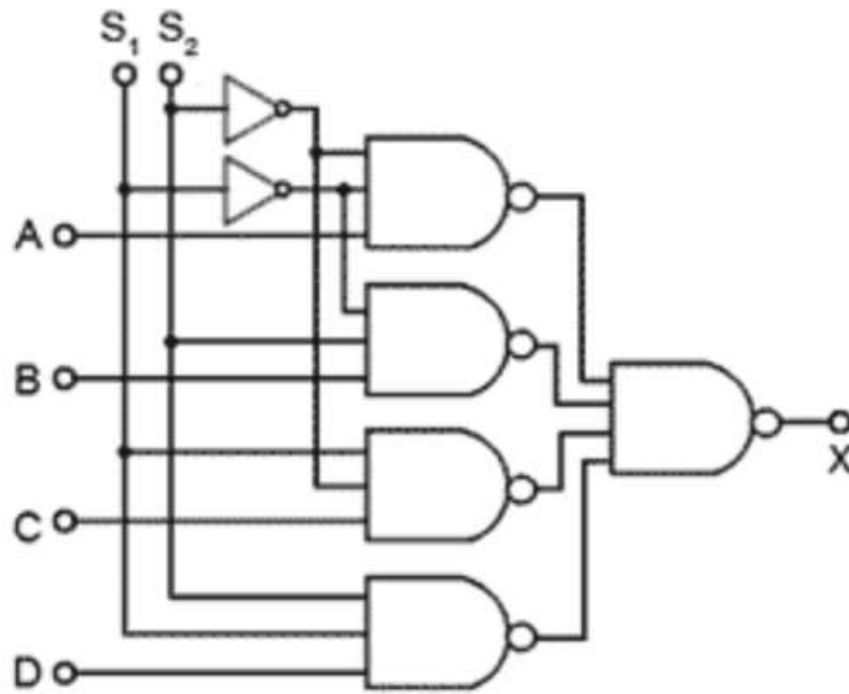
☐ a. 12

☒ b. 4

☐ c. 10

☐ d. 8

If $S_1=1$ and $S_2=0$ what will be the logic state at the output X?



☒ a. $X=C$

☐ b. $X=A$

☐ c. $X=D$

☐ d. $X=B$

Which of the following is the output 'Y3,Y2,Y1,Y0' for 2 to 4 decoder with enable pin "0" and inputs A,B is 01?

☒ 0000

☐ 0001

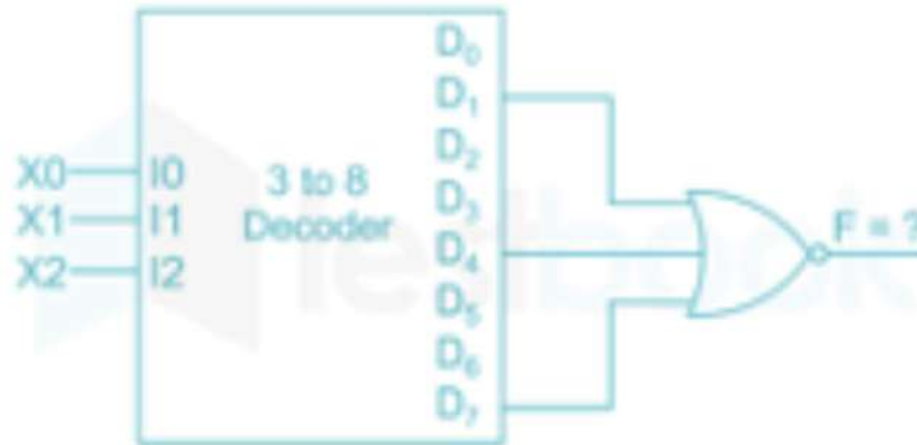
☐ 0010

☐ 0100

How many input and output required for multiplexer ?

- ☐ a. One input and many output
- ☐ b. None of the above
- ☐ c. One input and one output
- ☒ d. Many input and one output

A 3x8 decoder is shown below. Find $F(x_2, x_1, x_0)$



- ☐ $\sum m(1, 4, 7)$
- ☒ $\sum m(0, 2, 3, 5, 6)$
- ☐ $x_0 + x_1 + x_2$
- ☐ $\overline{x_0 + x_1 + x_2}$

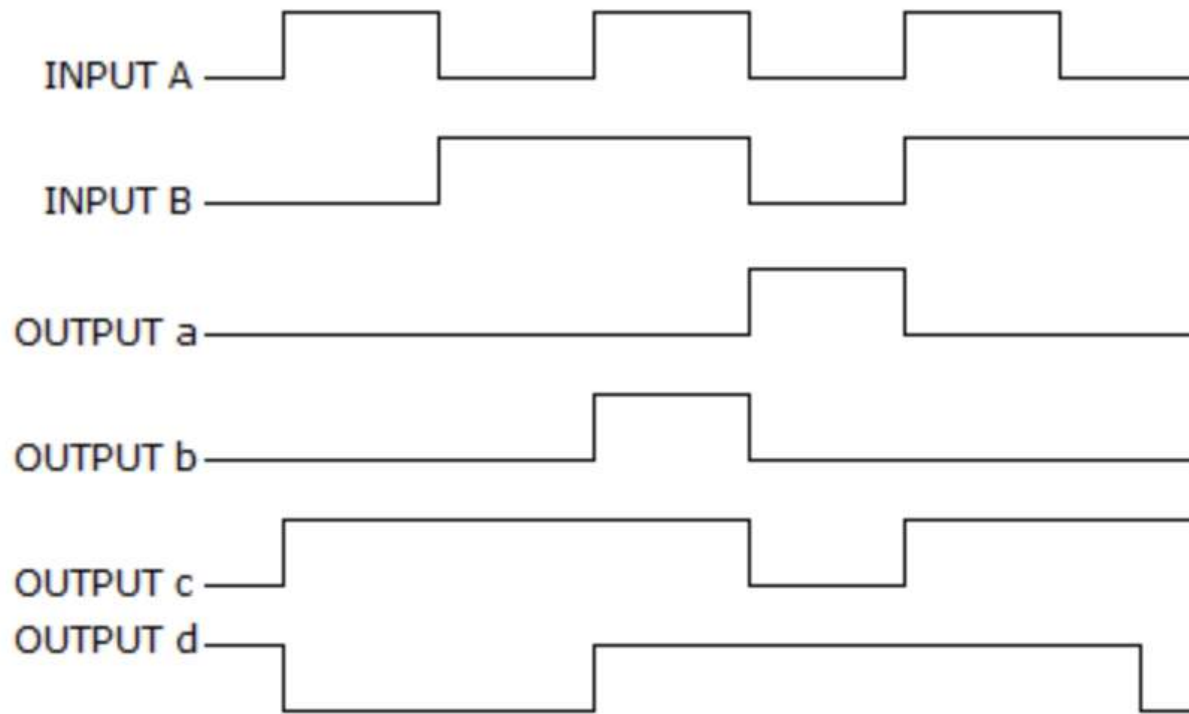
The expression $Y(A, B, C) = \sum m(1, 3, 5, 6)$ is to realized using a multiplexer. Then

- ☒ use 8 : 1 multiplexer and ground input lines 0, 2, 4, 7
- ☐ use 8 : 1 multiplexer and ground input lines 1, 3, 5, 6
- ☐ use 8 : 1 multiplexer and ground input lines 0, 1, 2, 3
- ☐ use 8 : 1 multiplexer and ground input lines 4, 5, 6, 7

A 32 decoder circuit can be implemented using:

- ☒ One 2x4 decoder and four 3x8 decoders
- ☐ Four 3x8 decoders
- ☐ Two 3x8 decoders
- ☐ Eight 2x4 decoders

For a two-input XNOR gate, with the input waveforms as shown below, which output waveform is correct?



☐ a. b

☐ b. a

☒ c. d

☐ d. c